# Selecting Nanomaterials for Enhanced Carbon Fiber Reinforced Polyamide Composites: A Strategic Framework

## Abstract

The integration of nanomaterials into carbon fiber reinforced polyamide (CF/PA) composites offers a promising route to overcome the inherent limitations of thermoplastic composites, particularly their poor interfacial adhesion and limited multifunctionality. This paper presents a systematic framework for selecting appropriate nanomaterials based on targeted property enhancement, interfacial compatibility, and processing feasibility. Key nanomaterial candidates are evaluated, including graphene nanoplatelets (GNPs), silica nanoparticles, cellulose nanofibrils (CNF), and zinc oxide (ZnO) nanostructures. Recent experimental evidence demonstrates that dual nanofiller strategies can yield synergistic improvements: silane-treated silica and graphene combined in CF/PA blends achieve a 32.3% increase in interlaminar shear strength (ILSS) and 33.2% improvement in impact strength . Cellulose nanofibrils applied as a sustainable sizing agent increase ILSS by 50% . ZnO nanorod growth combined with CNT-modified silane coating on carbon fibers produces up to 32% and 41% increases in tensile modulus and strength, respectively . A decision framework is proposed to guide nanomaterial selection based on the specific property targets, matrix chemistry, and manufacturing constraints of the CF/PA composite system.

**Keywords**: Carbon fiber; Polyamide composites; Nanomaterials; Interfacial adhesion; Mechanical properties

## 1. Introduction

Carbon fiber reinforced thermoplastic composites, particularly those utilizing polyamide (PA) matrices, have emerged as attractive alternatives to traditional thermoset composites due to their recyclability, rapid processing cycles, and high toughness . However, the non-polar surface of carbon fibers results in relatively weak interfacial bonding with polyamide matrices, limiting the full potential of these composites . This interfacial weakness manifests as reduced interlaminar shear strength, poor impact resistance, and suboptimal stress transfer between fiber and matrix.

Nanomaterial modification has emerged as a powerful strategy to address these interfacial challenges while simultaneously imparting additional functional properties such as thermal conductivity, electromagnetic interference shielding, and electrical conductivity . The selection of appropriate nanomaterials, however, is not straightforward: the vast array of available nanostructures, each with distinct morphology, surface chemistry, and physical properties, requires careful matching with the target polyamide matrix and processing conditions.

This paper establishes a systematic selection framework for nanomaterials in CF/PA composites. The framework is grounded in three primary selection criteria: (1) interfacial compatibility with both carbon fiber and polyamide, (2) targeted property enhancement capability, and (3) processing feasibility within typical thermoplastic composite manufacturing routes.

## 2. Nanomaterial Selection Criteria

### 2.1 Interfacial Compatibility

The fundamental challenge in CF/PA composites is achieving strong interfacial adhesion between the non-polar carbon fiber surface and the polyamide matrix. Effective nanomaterials must either bridge this chemical mismatch or create mechanical interlocking structures .

**Surface chemistry compatibility**: Nanomaterials with functional groups capable of hydrogen bonding or chemical reaction with polyamide amide groups are preferred. Silane coupling agents have proven particularly effective, acting as molecular bridges that covalently bond to both the nanomaterial/fiber surface and the polyamide matrix . Silane treatment of silica and graphene nanofillers fostered superior nanofiller-matrix bonding in CF/PA blends, contributing to the significant mechanical improvements observed .

**Mechanical interlocking**: Nanostructures that increase surface roughness and create physical entanglement with the polyamide chains can enhance interfacial adhesion even without strong chemical bonding. Zinc oxide nanorods grown on carbon fiber surfaces form mechanical

interlocking with the polymer matrix . ZnO nanoparticles with porous hollow structures provide embedding channels for molten nylon molecular chains during melt blending, realizing mechanical interlocking .

### 2.2 Property Enhancement Targets

Different nanomaterials offer distinct property profiles, and selection should be guided by the specific performance requirements of the target application.

**Mechanical reinforcement**: Graphene nanoplatelets (GNPs) and carbon nanotubes (CNTs) offer exceptional mechanical properties and high aspect ratios, making them effective for load transfer enhancement . GNPs possess high mechanical strength and high surface area, enabling better stress transfer between polymer and filler compared to CNTs, although their platelet structure presents agglomeration challenges .

**Thermal conductivity**: Graphene and its derivatives are particularly effective for enhancing thermal conductivity. Functionalized graphene nanoplatelets incorporated at 1 wt% significantly improved the dispersion and interfacial bonding in polyamide-6-based carbon fiber composites . Boron nitride nanosheets offer an electrically insulating alternative for applications requiring thermal management without electrical conductivity.

**Impact toughness**: Silica nanoparticles and graphene can synergistically improve impact strength. In CF/PA blends, the combination of silane-treated silica and graphene yielded a 33.2% increase in impact strength compared to the unfilled blend .

**Multifunctional properties**: $MnO_2$ nanosheets grown in situ on carbon fiber surfaces have been demonstrated to simultaneously enhance interfacial mechanical properties and electromagnetic interference shielding in CF/PA6 composites . ZnO nanostructures can impart antibacterial and UV-blocking functionality while enhancing mechanical properties.

### 2.3 Processing Feasibility

The selection of nanomaterials must consider compatibility with standard thermoplastic composite manufacturing processes, including extrusion, injection molding, and compression molding.

**Dispersion challenges**: Nanomaterial agglomeration is a persistent challenge, particularly for high-aspect-ratio fillers like GNPs. The van der Waals interactions between graphene layers lead to agglomerate formation, reducing the effective surface area and deteriorating reinforcement performance . Surface treatment with silane coupling agents can mitigate agglomeration while improving matrix compatibility.

**Thermal stability**: The nanomaterial must withstand polyamide processing temperatures (typically 240–300°C for PA6 and PA66). Most inorganic nanomaterials including silica, ZnO, and carbon-based fillers are thermally stable within this range.

**Sustainability considerations**: Cellulose nanofibrils (CNF) represent an emerging sustainable option, derived from renewable resources and offering excellent mechanical properties. CNF sizing on commingled CF/PA6 fibers increased ILSS by 50% while maintaining thermal stability .

## 3. Nanomaterial Candidates: Comparative Evaluation

### 3.1 Graphene Nanoplatelets (GNPs)

GNPs are two-dimensional carbon nanostructures with high specific surface area and excellent mechanical properties. Their platelet morphology provides better stress transfer than one-dimensional CNTs . However, agglomeration remains the primary limitation.

**Performance data**: In CF/PA blends, functionalized graphene at 1 wt% significantly improved dispersion and interfacial bonding . Combined with silica in a dual-filler strategy, graphene contributed to a 32.3% ILSS increase .

**Selection rationale**: Optimal for applications requiring combined mechanical reinforcement and thermal/electrical conductivity enhancement.

### 3.2 Silica Nanoparticles

Silica nanoparticles offer excellent compatibility with polyamide matrices through silanol chemistry and can be easily functionalized with silane coupling agents.

**Performance data**: Silane-treated silica at 1.5 wt% as sole reinforcement yielded modest improvements (8.5% hardness, 5.6% tensile strength) . However, when combined with graphene, the synergistic effect produced substantially greater enhancements (14.1% hardness, 16% tensile strength) .

**Selection rationale**: Excellent for synergistic combinations with carbon-based nanofillers and for applications where cost-effectiveness is prioritized.

### 3.3 Cellulose Nanofibrils (CNF)

CNF represent a sustainable, renewable nanomaterial option with high aspect ratio and excellent mechanical properties.

**Performance data**: CNF sizing on commingled CF/PA6 yarns increased ILSS by 50%, flexural strength by 11%, and tensile strength by 2.5% . The improvement was attributed to enhanced interfacial bonding between CF and PA6.

**Selection rationale**: Ideal for sustainable composite applications and as a sizing agent for commingled fiber architectures.

### 3.4 Zinc Oxide (ZnO) Nanostructures

ZnO nanostructures offer unique multifunctional capabilities including antibacterial activity, UV blocking, and the ability to form mechanical interlocking with polymer matrices .

**Performance data**: ZnO nanorods grown on carbon fiber surfaces combined with CNT-modified silane coating produced up to 32% and 41% increases in tensile modulus and strength, respectively, and 107% improvement in in-plane shear strength . ZnO nanoparticles with porous hollow structures improved tensile strength, surface hardness, and wear resistance of nylon-based composites .

**Selection rationale**: Optimal for multifunctional applications requiring mechanical enhancement combined with antibacterial or UV-blocking properties.

### 3.5 Comparative Summary

| Nanomaterial | Primary Benefit | Key Challenge | Best Application |

|:---|:---|:---|:---|

| Graphene (GNP) | Thermal/electrical conductivity, mechanical | Agglomeration | Multifunctional structural |

| Silica | Cost-effective, synergistic with carbon fillers | Modest standalone effect | Dual-filler systems |

| CNF | Sustainable, excellent sizing agent | Thermal stability limit | Eco-friendly composites |

| ZnO | Multifunctional (antibacterial, UV) | Dispersion control | Functional structural |

## 4. Dual Nanofiller Strategy

The most significant recent advancement in CF/PA nanocomposite design is the recognition that single nanofillers often cannot simultaneously address all performance requirements. Dual nanofiller strategies exploit synergistic effects between complementary nanomaterials .

The combination of silane-treated silica and graphene in CF/PA blends exemplifies this approach. While silica alone at 1.5 wt% produced modest improvements, the silica-graphene combination yielded dramatically superior results through complementary mechanisms: silica improved nanofiller-matrix bonding and dispersion, while graphene provided high aspect ratio reinforcement and network formation .

This synergistic effect resulted in:

- 32.3% increase in interlaminar shear strength
- 14.1% increase in hardness
- 16% increase in tensile strength
- 19% increase in flexural strength
- 33.2% increase in impact strength

The mechanism involves improved network structure formation and enhanced dispersion of both fillers when used together, compared to their individual use .

## 5. Selection Framework and Decision Criteria

Based on the analysis above, a systematic framework for nanomaterial selection in CF/PA composites is proposed:

**Step 1: Define primary property targets.** Identify whether the application prioritizes mechanical reinforcement, thermal management, impact toughness, or multifunctional performance.

**Step 2: Assess interfacial compatibility.** Evaluate the nanomaterial's surface chemistry against the specific polyamide grade (PA6, PA66, PA12) and determine whether surface functionalization (silane treatment, plasma activation) is required.

**Step 3: Consider processing route.** Match the nanomaterial to the intended manufacturing process (extrusion, injection molding, compression molding, in-situ polymerization) and assess dispersion feasibility.

**Step 4: Evaluate synergistic potential.** For multifunctional requirements, consider dual or hybrid nanofiller strategies where complementary mechanisms can operate simultaneously.

**Step 5: Validate through prototype testing.** Conduct mechanical (tensile, flexural, ILSS, impact) and functional (thermal conductivity, EMI shielding) characterization to confirm performance gains.

## 6. Conclusions

The selection of nanomaterials for enhancing CF/PA composite properties requires careful consideration of interfacial compatibility, targeted property enhancement, and processing feasibility. Graphene nanoplatelets offer the broadest property profile but suffer from agglomeration challenges. Silica nanoparticles are cost-effective and excel in synergistic combinations. Cellulose nanofibrils provide a sustainable option with demonstrated interfacial benefits. Zinc oxide nanostructures enable multifunctional performance beyond mechanical reinforcement.

The dual nanofiller strategy represents the state of the art, with silane-treated silica and graphene demonstrating synergistic improvements of 32.3% in ILSS and 33.2% in impact strength. Future research should focus on scalable surface treatment methods, understanding the mechanisms of nanofiller synergy, and developing multifunctional nanomaterial systems that address multiple performance requirements simultaneously.

## References

[1] X. Yang, X. Li, S. Zhou, N. Song, T. Xu, P. Ding, Tailoring interfaces and networks in functionalized graphene-carbon fiber reinforced polyamide composites for enhanced thermal and mechanical properties, Polym. Compos. 46 (2025) 15419–15432.

[2] S. Suresha, et al., Synergetic effect of graphene nanoplatelet, carbon fiber and coupling agent addition on the tribological, mechanical and thermal properties of polyamide 6,6 composites, Composites Part B: Engineering (2019).

[3] S. Srinivas, H. Hema, D.C. Madaiah, R. Hemanth, B. Suresha, I. Siva, S.S. Dol, K.A. Ahmad, M.T.H. Sultan, Dual nanofiller strategy for enhanced mechanical properties in carbon fiber reinforced polyamide blends, Nanotechnol. Rev. 15 (2026) 20250235.

[4] U. Marathe, H. Tekinalp, G. Chahine, et al., Sizing comingled CF/PA 6 fibers with cellulose nanofibrils for enhanced performance properties, Composites Part A 200 (2026) 109308.

[5] A. Kausar, Advances in carbon fiber reinforced polyamide-based composite materials, Adv. Mater. Sci. 19 (2019) 67–82.

[6] In-situ polymerized polyamide-6 and ZnO nanorod-grown woven carbon fiber coated with CNT-modified silane, Compos. Sci. Technol. (2018).

[7] Mussel-like carbon fiber/$MnO_2$ nanosheet heterostructures for mechanically strong carbon fiber/polyamide composites with excellent electromagnetic interference shielding, Composites Part B (2024).

[8] Nanoparticle-reinforced nylon-based engineering plastic, Chinese Patent CN122080627A.

[9] M. Arnold, M. Henne, K. Bender, K. Drechsler, Polyamide 12 modified with nanoparticles: Effect on impact behaviour and on the electrical conductivity of carbon fibre-reinforced epoxy composites, J. Compos. Mater. 50 (2016) 159–171.